Vorlesung: Grundlagen der Computergrafik

Kapitel 4: WebGL und Three.js

Kapitel 4.3:

Textures in Three.js

Vorlesung im Sommersemester 2026
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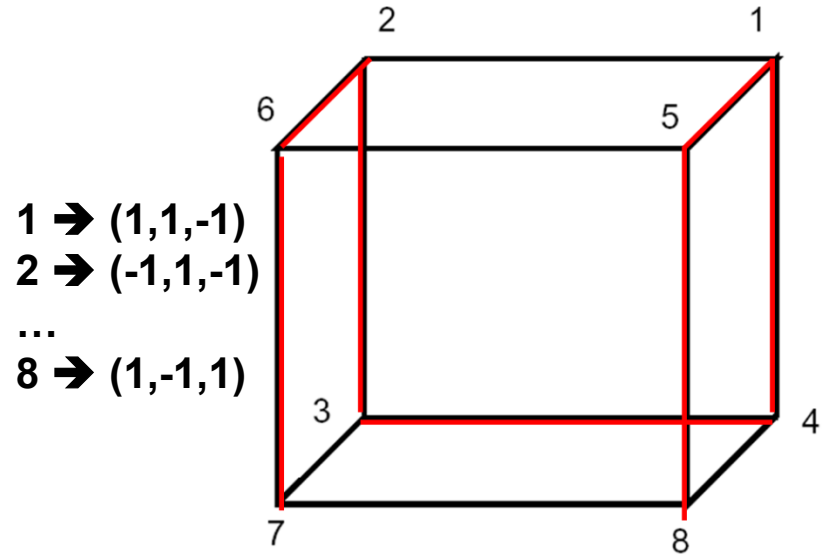


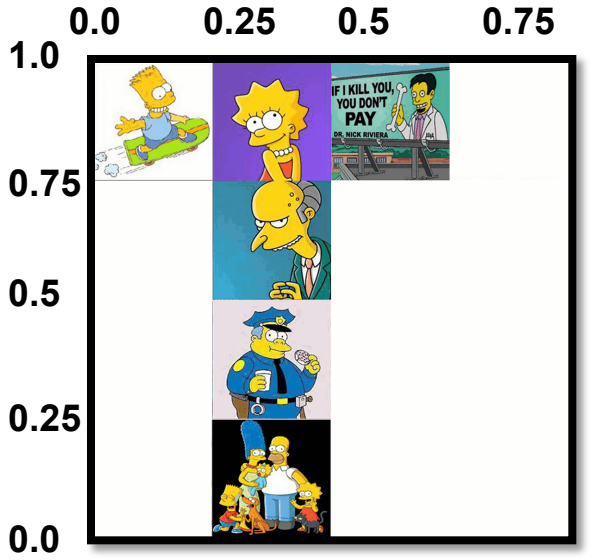
TEXTURE MAPPING MIT THREE.MESHBASICMATERIAL

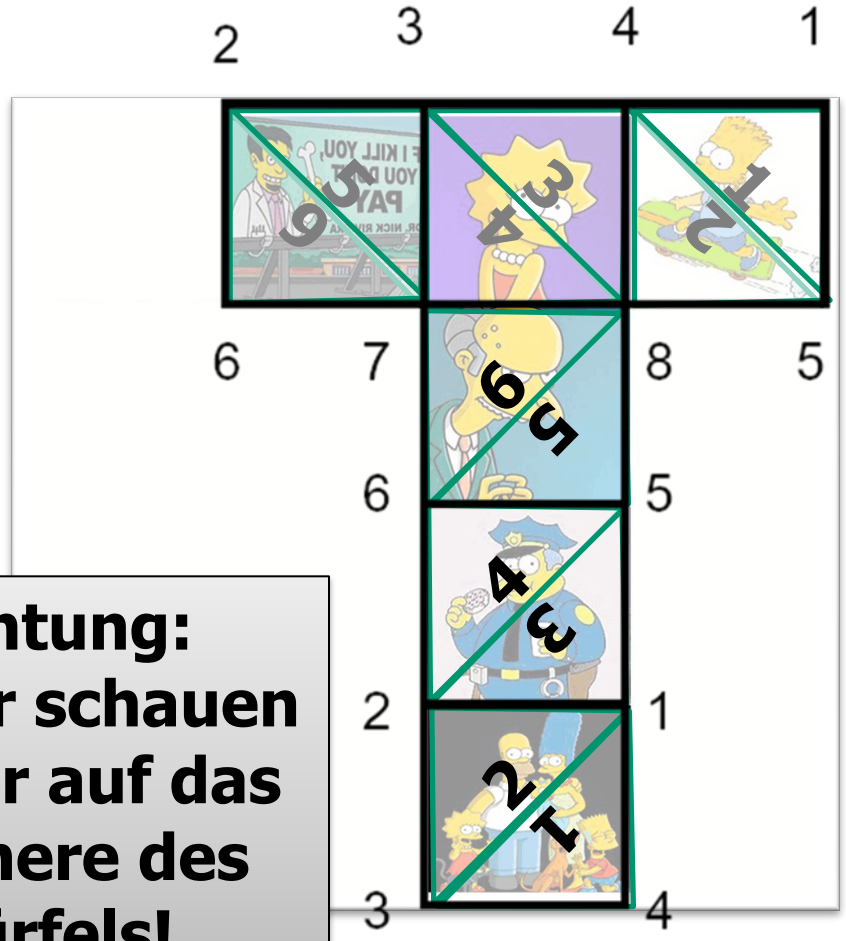
Texture Mapping in Three.js

```
// The TextureLoader loads the texture
// and then completes the scene
var textureLoader = new THREE.TextureLoader();
var texture = textureLoader.load('simpsons.gif', function(mymap) {
    // Now, create a basic material; pass in the map
    var material = new THREE.MeshBasicMaterial({ map: mymap });
    // Create the cube geometry and the UV-mapping.
    var geometry = new THREE.Geometry();
    // Add vertices of the geometry
    geometry.vertices.push(new THREE.Vector3( 1.0,  1.0, -1.0));
    geometry.vertices.push(new THREE.Vector3(-1.0,  1.0, -1.0));
    …
    // The faceVertexUvs property of geometry is an array of arrays
    // that contains the coordinate mapping for each face of the geometry
    geometry.faceVertexUvs[0]=[]
    // Create UV-Coordinates
    var uv_a=new THREE.Vector2(0, 1);
    var uv_b=new THREE.Vector2(0.25, 1);
    …
    // Add faces and the UV-mapping for each face
    geometry.faces.push(new THREE.Face3(0, 4, 3));
    geometry.faceVertexUvs[0].push( [uv_a, uv_e, uv_b] );
    geometry.faces.push(new THREE.Face3(7,3,4));
    geometry.faceVertexUvs[0].push( [uv_f, uv_b, uv_e] );
    …
    // And put the geometry and material together into a mesh
    cube = new THREE.Mesh(geometry,material);
    …
})
```

Würfel als Triangle Strip mit Texturen

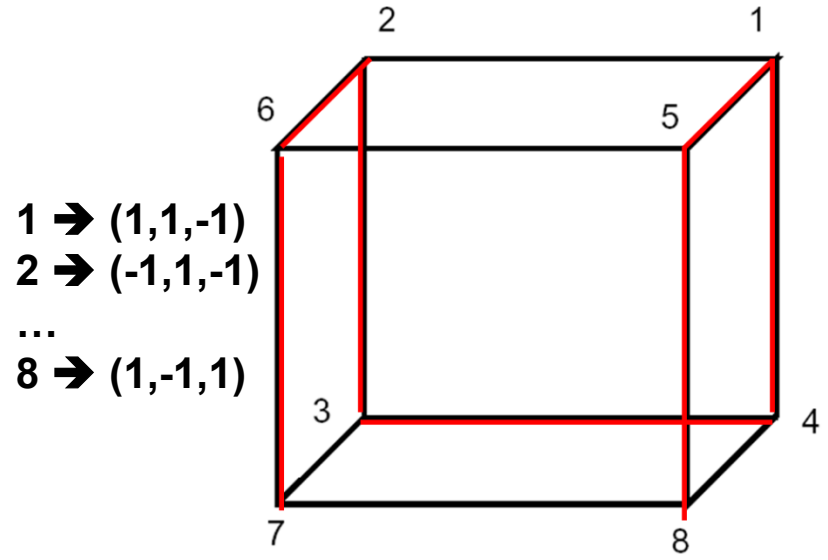


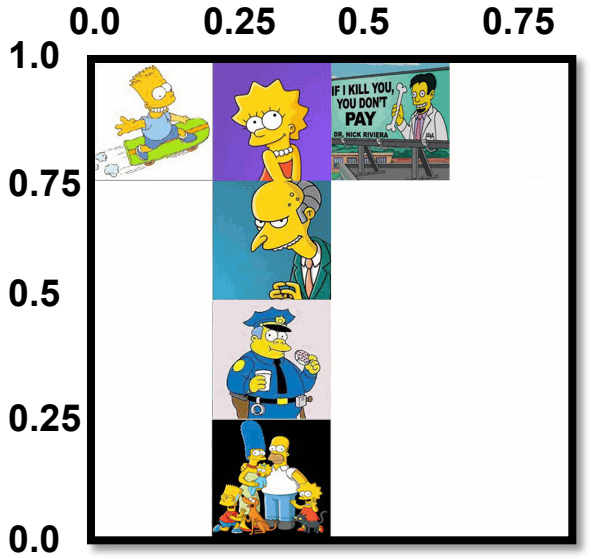


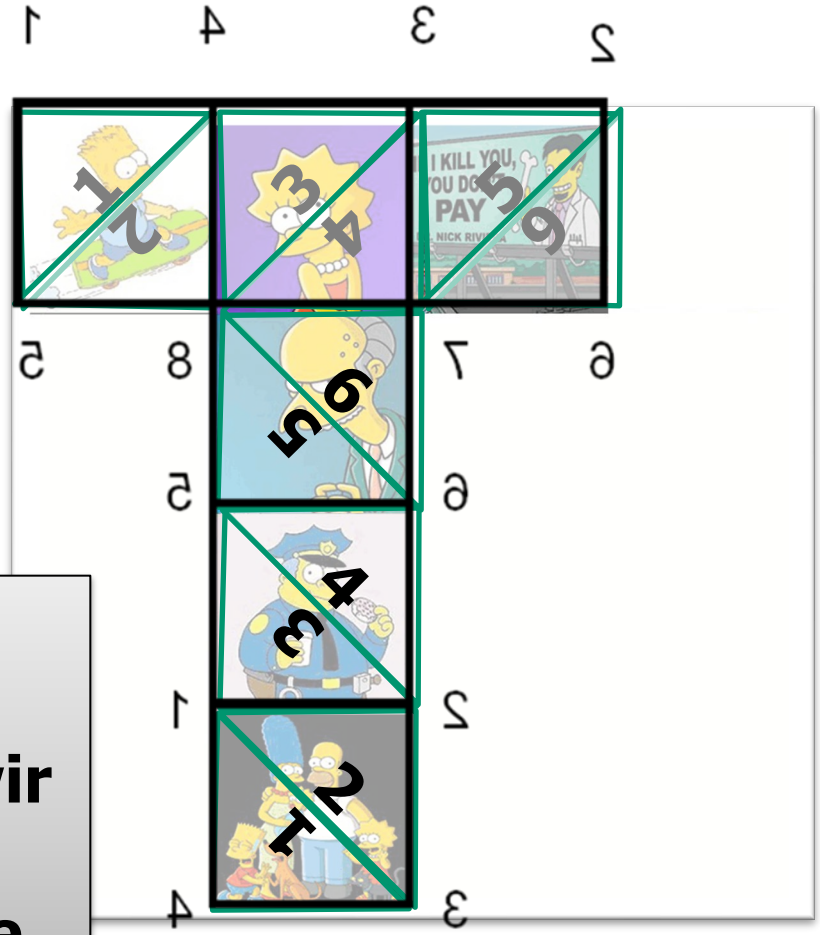


Achtung: Wir schauen hier auf das Innere des Würfels!

Würfel als Triangle Strip mit Texturen



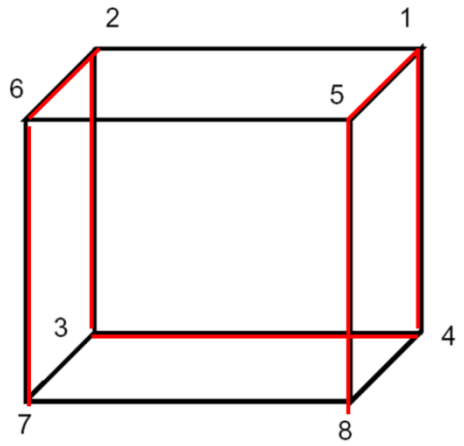


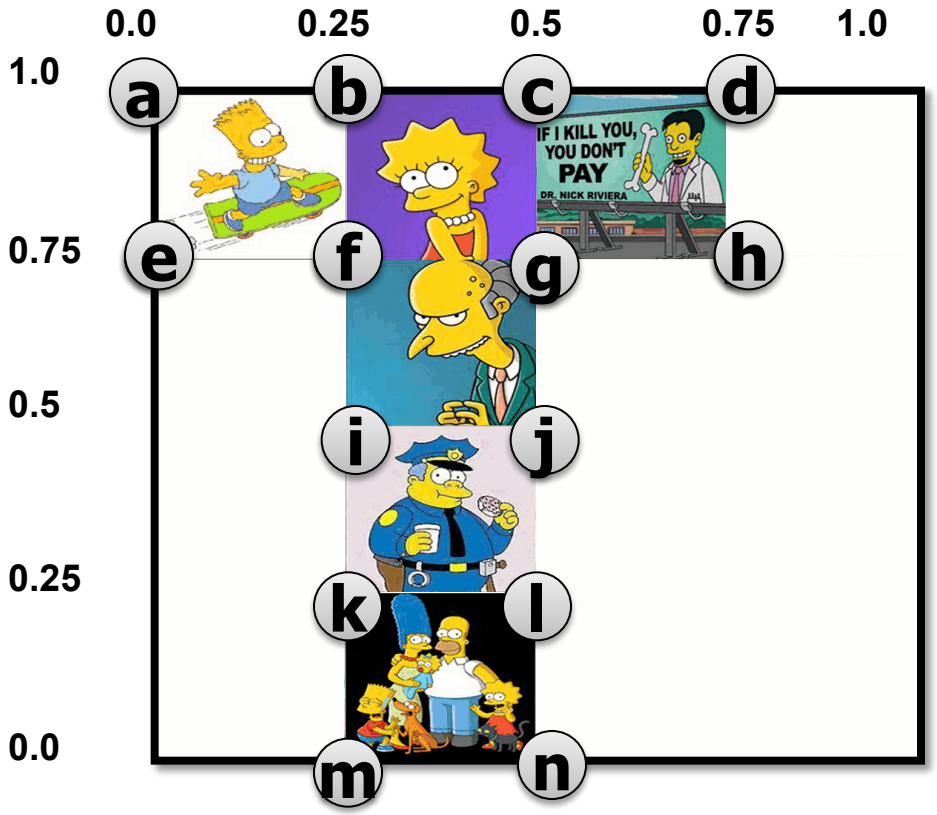


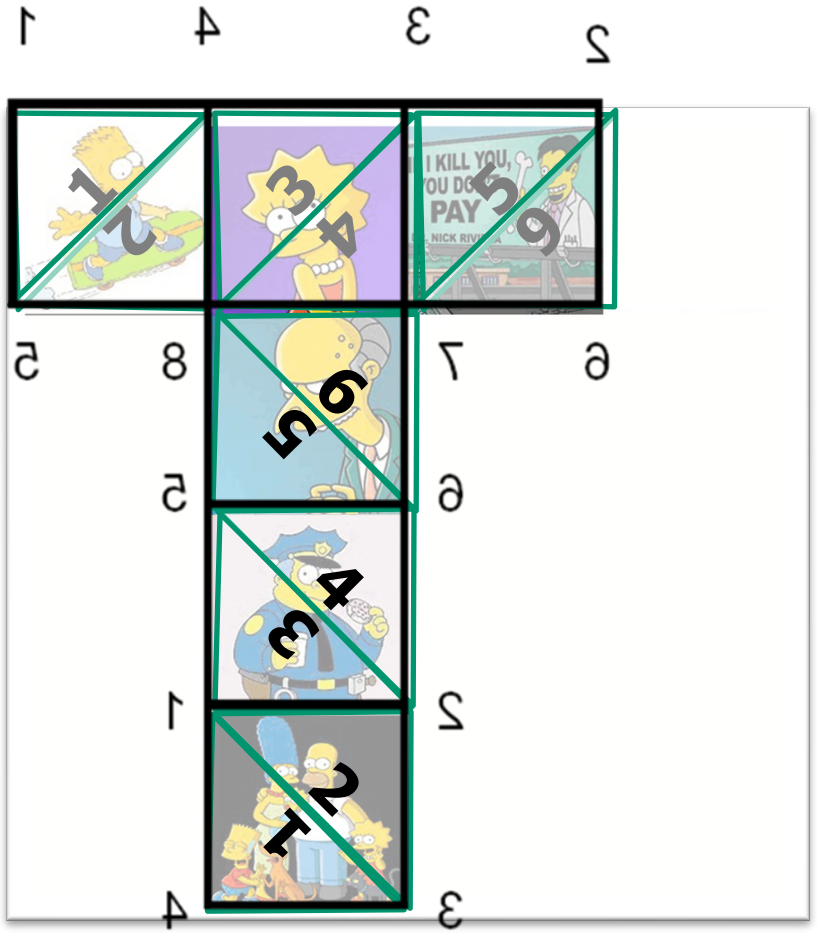
Achtung: Jetzt schauen wir auf die Außenseite

Datei: textures.html

1 ➔ (1,1,-1)
2 ➔ (-1,1,-1)
…
8 ➔ (1,-1,1)







Aufgabe 1

- Mappen Sie die Simpsons auf den Würfel.
- Datei:

https://www.st.uni-trier.de/lectures/S26/CG/webgl/textures/textures.html

- Im Editor:

https://www.st.uni-trier.de/lectures/S26/CG/webgl/textures/textures_ace.html

- Alles in einer ZIP-Datei:

https://www.st.uni-trier.de/lectures/S26/CG/webgl/textures.zip

http://www.shaderific.com/

TEXTURE MAPPING MIT THREE.SHADERMATERIAL

Texture Mapping mit Shadern

```
// The TextureLoader loads the texture
// and then completes the scene
var textureLoader = new THREE.TextureLoader();
var texture = textureLoader.load('simpsons.gif', function(texture) {
    // Now, create a shader material; pass in the map
    var material = new THREE.ShaderMaterial({
        vertexColors: THREE.VertexColors,
        uniforms : { myTexture: { type: "t", value: texture },
                     aspectRatio: { type: "f",
                                    value: ( container.offsetWidth /
                                             container.offsetHeight) } },
        vertexShader: document.getElementById( 'tx-shader-vs' ).textContent,
        fragmentShader: document.getElementById( 'tx-shader-fs' ).textContent
      });
    …
  }
```

```
<script id="tx-shader-fs"
        type="x-shader/x-fragment">
 varying vec3 vColor;
 varying vec2 vUv;
 varying vec3 vertex_normal;
 uniform sampler2D myTexture;

 void main(void) {
  vec4 col = texture2D(myTexture, vUv);
  gl_FragColor = col;
 }
</script>
```

```
<script id="tx-shader-vs"
        type="x-shader/x-vertex">
 varying vec3 vColor;
 varying vec2 vUv;
 varying vec3 vertex_normal;

 void main(void) {
  vColor = color;
  vUv = uv;
  vertex_normal = normalMatrix*normal;
  gl_Position = projectionMatrix
   *modelViewMatrix*vec4(position, 1.0);
 }
</script>
```